

From: Sarah Johnson – Owner, Urban Threads Clothing
To: Astrax Analytics Solutions – Data Engineering & Analytics Partner
Project Lead: Samuel Sichone — Director, Astrax Analytics Solutions
Subject: Development of a Customer Shopping Behavior Analytics Platform

1. Business Background

Urban Threads Clothing is a growing fashion retail business offering clothing and accessories through both its physical store and online sales channels. As the business expands, management has observed changing customer purchasing patterns across different demographics, product categories, seasons, and sales channels. However, the company currently relies on spreadsheets and disconnected reports, making it difficult to gain meaningful insights into customer behavior and sales performance.

The lack of a centralized analytics platform limits the company's ability to understand purchasing trends, evaluate marketing effectiveness, identify loyal customers, and optimize inventory planning. Urban Threads seeks a modern data analytics solution that transforms raw sales data into actionable business intelligence for strategic decision-making.

2. Business Objectives

The proposed solution should enable Urban Threads to:

- Understand customer shopping behavior across different demographics.
 - Identify purchasing trends by product category, season, and sales channel.
 - Measure the impact of discounts, customer reviews, and promotions on sales.
 - Analyze repeat purchases and customer loyalty.
 - Provide management with interactive dashboards for informed business decisions.
 - Automate monthly reporting to reduce manual analysis efforts.
-

3. Functional Requirements

3.1 Data Sources

The platform will process retail data collected from multiple operational sources, including:

- Customer transaction records (CSV files)
 - Product catalog and inventory information
 - Customer demographic information
 - Sales channel data (Online and In-Store)
 - Payment method information
 - Customer ratings and product reviews
-

3.2 Data Processing & Storage

The solution will implement a structured ETL pipeline consisting of:

Data Ingestion

- Import raw CSV datasets using Python.
- Validate incoming data before processing.
- Standardize column names and data formats.

Data Cleaning & Transformation

Python will be used to resolve common retail data quality issues including:

- Missing customer information
- Duplicate transaction records
- Invalid payment methods
- Incorrect date formats
- Inconsistent product categories
- Missing ratings and review values

Database Storage

Following transformation, the cleaned dataset will be loaded into a relational SQL database designed using a Star Schema consisting of:

Fact Tables

- Sales Transactions

Dimension Tables

- Customers
- Products
- Dates
- Sales Channels
- Payment Methods

The database will support efficient analytical querying and reporting.

3.3 Analytics

SQL will be used to generate business insights including:

- Customer segmentation by age and gender
- Best-selling product categories
- Monthly and seasonal sales trends
- Online versus in-store sales performance
- Repeat customer analysis
- Customer loyalty metrics
- Average order value
- Impact of discounts on purchasing behavior
- Most preferred payment methods
- Customer review and rating analysis

Power BI dashboards will present interactive KPIs including:

- Total Revenue
 - Total Orders
 - Customer Growth
 - Top Selling Products
 - Monthly Sales Trend
 - Sales by Gender
 - Sales by Age Group
 - Sales by Product Category
 - Sales Channel Performance
 - Payment Method Distribution
 - Customer Loyalty Overview
 - Discount Effectiveness
 - Geographic Sales Distribution (if location data is available)
-

3.4 Automation & Reporting

The solution will automate the analytics workflow by:

- Running scheduled Python ETL processes
 - Loading cleaned datasets into the SQL database
 - Refreshing Power BI dashboards automatically
 - Generating monthly management reports
 - Maintaining data validation logs for monitoring pipeline health
-

3.5 Data Quality

The platform will simulate and resolve realistic retail data issues including:

- Missing customer IDs
- Duplicate purchase transactions
- Missing product categories
- Invalid transaction dates
- Incorrect pricing values
- Missing payment methods
- Inconsistent gender values
- Null customer ratings

All validation and cleansing will occur during the Python ETL process before data is loaded into the SQL database.

3.6 Security

The solution will implement appropriate security measures including:

- Role-Based Access Control (RBAC) for management reporting

- Secure database authentication
 - Read-only dashboard access for business users
 - Regular database backups
 - Version-controlled source code using GitHub
-

4. Deliverables

The completed project will include:

- Python ETL pipeline for data ingestion, cleaning, and transformation.
 - Structured SQL database with Star Schema data model.
 - SQL scripts for business analysis and reporting.
 - Interactive Power BI dashboard connected to the SQL database.
 - Automated monthly reporting workflow.
 - Complete technical documentation including:
 - System Architecture
 - Data Model
 - ETL Process Documentation
 - Database Design
 - User Guide
 - Well-organized GitHub repository containing:
 - Python scripts
 - SQL queries
 - Database scripts
 - Power BI dashboard files
 - Project documentation
-

5. Success Criteria

The project will be considered successful when:

- Raw retail CSV data is automatically cleaned and transformed using Python.
- Data is successfully stored within a structured SQL database.
- SQL queries generate accurate customer and sales insights.
- Power BI dashboards refresh using the latest processed data.
- Monthly reports are generated with minimal manual intervention.
- Business stakeholders can monitor customer behavior through interactive dashboards.
- Data quality issues are automatically detected and resolved during processing.
- All project components are documented and maintained within a GitHub repository.